

LIOUVILLE-TYPE THEOREMS FOR COUPLED-DRIFT MONGE-AMPÈRE EQUATIONS

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ABSTRACT. In this paper, we study entire solutions and periodic correctors for the coupled-drift Monge–Ampère equation

$$\det D^2u = \exp\{-a \cdot Du + b \cdot x + V(x) - c_0\}, \quad D^2u > 0.$$

For $V \equiv 0$, we obtain a sharp classification of the whole-space solvability regimes: all entire smooth strictly convex solutions are quadratic when $a = b = 0$; no such solution exists when $a \neq 0$ and $a \cdot b \leq 0$; and non-quadratic entire solutions exist when $a = 0$ and $b \neq 0$, or when $a \cdot b > 0$. For the null case $a \neq 0$, $a \cdot b = 0$, we give a scalar maximum-principle argument in every dimension $n \geq 2$.

For periodic V , we prove existence and uniqueness of the normalized pair (ψ_A, c_A) solving the drifted cell problem

$$\det(A + D^2\psi) = \exp\{-a \cdot D\psi + V - c_A\}, \quad A + D^2\psi > 0 \quad \text{on } \mathbb{T}^n.$$

We also prove that any asymptotically quadratic entire solution must satisfy $b = Aa$. If its remainder is bounded, then the solution is the corresponding quadratic-periodic corrector up to an additive constant.

1. INTRODUCTION

In this paper, we study entire smooth strictly convex solutions of the Monge–Ampère equation

$$(1.1) \quad \det D^2u = \exp\{-a \cdot Du + b \cdot x + V(x) - c_0\}, \quad D^2u > 0, \quad x \in \mathbb{R}^n,$$

where $a, b \in \mathbb{R}^n$, $c_0 \in \mathbb{R}$, and V is either zero or a smooth periodic function on $\mathbb{T}^n$. When $V \equiv 0$, the Legendre transform interchanges the two drift terms. Indeed, if $y = Du(x)$ and $f = u^*$, then on the gradient image $Du(\mathbb{R}^n)$,

$$\det D^2f = \exp\{a \cdot y - b \cdot Df + c_0\}.$$

For the whole-space problem, the sign of $a \cdot b$ determines the solvability. For periodic correctors, the corresponding condition is $b = Aa$, where A is the quadratic part of the solution.

We first recall some classical results for the real Monge–Ampère equation. The Jörgens–Calabi–Pogorelov theorem states that every entire smooth strictly convex solution of

$$\det D^2u = \text{constant} \quad \text{in } \mathbb{R}^n$$

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is a quadratic polynomial; see [Jör54, Cal58, Pog72]. Cheng and Yau [CY86] gave an analytic proof in their study of complete affine hypersurfaces. Caffarelli and Li [CL03] extended the theory to viscosity solutions and exterior-domain problems. Entire equations with nonconstant prescribed densities were also studied by Chou and Wang [CW96]. We refer to [Gut16, Fig17, Le24] for further results on the Monge–Ampère equation and its linearized equation.

A related equation arises in affine Kähler–Ricci flat geometry. After a Legendre transform, it takes the form

$$\det D^2u = \exp\{-d \cdot Du - d_0\} \quad \text{in } \mathbb{R}^n.$$

Li and Xu [LX09] proved that there is no entire smooth strictly convex solution when $d \neq 0$, and Xu and Zhu [XZ16] later gave a shorter proof. For related affine Bernstein problems, see [LXSJ10, TW00, TW02, TW05, TW08].

We first consider the constant-medium case of (1.1), namely

$$(1.2) \quad \det D^2u = \exp\{-a \cdot Du + b \cdot x - c_0\}, \quad D^2u > 0, \quad x \in \mathbb{R}^n.$$

This equation also appears in the study of Lagrangian translating solitons in the pseudo-Euclidean space $\mathbb{R}_n^{2n}$; see Joyce–Lee–Tsui [JLT10] for related translating solitons in the Euclidean setting. Xu and Huang [XH13] derived the equation for space-like Lagrangian translating gradient graphs and proved rigidity under a completeness assumption. Further results were obtained in [HX15, XZ15, CQ16, XL19]. In particular, Xu and Zhu [XZ15] classified the one-dimensional entire solutions and obtained higher-dimensional rigidity under an asymptotic decay condition on the Hessian.

In the translating-soliton interpretation, the sign of $a \cdot b$ corresponds to the causal type of the translating vector, up to the convention for the null metric. Wu and Xu [WX21] proved nonexistence when $a \cdot b < 0$. In dimension two, Song–Wu–Xu [SWX26] proved nonexistence in the lightlike case when both drift vectors are nonzero. Our first result gives the complete solvability alternatives for (1.2).

Theorem 1.1. *Let $n \geq 1$, $a, b \in \mathbb{R}^n$, and $c_0 \in \mathbb{R}$. Consider (1.2). Then the following statements hold.*

- (i) *If $a = b = 0$, then every entire smooth strictly convex solution of (1.2) is a quadratic polynomial.*
- (ii) *If $a \neq 0$ and $a \cdot b \leq 0$, then (1.2) admits no entire smooth strictly convex solution.*
- (iii) *If either $a = 0$ and $b \neq 0$, or $a \cdot b > 0$, then (1.2) admits non-quadratic entire smooth strictly convex solutions.*

The new case in Theorem 1.1 is

$$a \neq 0, \quad a \cdot b = 0.$$

The main idea in the null case is as follows. We first prove a global oscillation bound for $w = a \cdot Du$. Setting

$$P = a^\top D^2u a,$$

the drifted linearized operator satisfies

$$\mathcal{L}w = 0, \quad \mathcal{L}P \geq 0,$$

with an explicit square identity for $\mathcal{L}P$. We then apply the maximum principle to $-P^{-\theta} - \varepsilon u$. This gives the contradiction for $n \geq 2$, while the case $n = 1$ follows directly from the ODE. No completeness assumption or decay condition on the Hessian is used.

We turn to periodic V . Periodic Monge–Ampère equations on flat or Hessian manifolds have been studied by Cheng and Yau [CY82], Li [Li90], Caffarelli–Viaclovsky [CV01], and Guedj–Tô [GT24]. For the undrifted equation with a positive periodic density, Caffarelli and Li [CL04] proved that an entire strictly convex solution has the form

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi(x),$$

where $A = A^\top > 0$ and ψ is periodic. Related results for less regular periodic data were obtained in [LL22, JLT25]. We also mention the recent work of Li and Liang [LL26] on quadratic-growth solutions of fully nonlinear elliptic equations with periodic data. For periodic homogenization, see [BLP78, Eva89, Eva92].

For a prescribed matrix $A = A^\top > 0$, the natural cell problem associated with (1.1) is

$$(1.3) \quad \det(A + D^2\psi) = \exp\{-a \cdot D\psi + V - c\}, \quad A + D^2\psi > 0 \quad \text{on } \mathbb{T}^n,$$

where the constant c is also unknown. The second result of this paper is the following.

Theorem 1.2. *For every $A = A^\top > 0$, $a \in \mathbb{R}^n$, and $V \in C^\infty(\mathbb{T}^n)$, there exists a unique pair (ψ_A, c_A) such that*

$$\psi_A \in C^\infty(\mathbb{T}^n), \quad \int_{\mathbb{T}^n} \psi_A = 0, \quad c_A \in \mathbb{R},$$

and

$$\det(A + D^2\psi_A) = \exp\{-a \cdot D\psi_A + V - c_A\}, \quad A + D^2\psi_A > 0 \quad \text{on } \mathbb{T}^n.$$

We prove the theorem by the continuity method. The bound $D^2\psi \geq -A$ and periodicity give uniform C^0 and C^1 estimates. For the C^2 estimate, the drift term in the full linearized operator cancels the third-order term arising from the largest eigenvalue. The higher-order estimates then follow from Evans–Krylov and Schauder theory.

The normalizing constant satisfies

$$\min_{\mathbb{T}^n} V - \log \det A \leq c_A \leq \max_{\mathbb{T}^n} V - \log \det A.$$

For fixed A and a , it is 1-Lipschitz with respect to V in L^∞ , and for fixed a and V , it is decreasing with respect to A in the Loewner order.

For comparison with weighted Monge–Ampère equations on compact Hessian manifolds, set

$$\Phi_A(x) = \frac{1}{2}x^\top Ax + \psi(x), \quad p(x) = D\Phi_A(x) = Ax + D\psi(x).$$

Then (1.3) can be written as

$$e^{a \cdot p(x)} \det Dp(x) = e^{a \cdot Ax + V(x) - c},$$

and

$$p(x + k) = p(x) + Ak, \quad k \in \mathbb{Z}^n.$$

This is related to the weighted Monge–Ampère equations studied by Hultgren and Önnheim [HÖ19]. Here, however, the source and target weights transform by the same character under lattice translations rather than being invariant, so their invariant-measure formulation does not apply directly.

The cell problem also gives the compatibility condition for quadratic-periodic solutions. If

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi(x) + C,$$

then the nonperiodic part of the exponent in (1.1) is $(b - Aa) \cdot x$. Hence such a solution can exist only when

$$b = Aa.$$

The same condition is still necessary under a subquadratic asymptotic assumption. For prescribed densities converging to periodic data, related quadratic-periodic asymptotics were obtained by Teixeira–Zhang [TZ16] and, more recently, by Qi–Bao [QB25].

Theorem 1.3. *Let $V \in C^\infty(\mathbb{T}^n)$, and let $u \in C^\infty(\mathbb{R}^n)$ be a strictly convex solution of*

$$\det D^2u = \exp\{-a \cdot Du + b \cdot x + V(x) - c_0\} \quad \text{in } \mathbb{R}^n.$$

Suppose that, for some $A = A^\top > 0$ and $\ell \in \mathbb{R}^n$,

$$f(x) = u(x) - \frac{1}{2}x^\top Ax - \ell \cdot x$$

satisfies

$$\sup_{B_R} |f| = o(R^2) \quad \text{as } R \rightarrow \infty.$$

Then $b = Aa$.

If, in addition, $f \in L^\infty(\mathbb{R}^n)$, then

$$c_0 + a \cdot \ell = c_A$$

and

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi_A(x) + C$$

for some $C \in \mathbb{R}$.

For the first conclusion we compare the exponential growth caused by $b - Aa \neq 0$ with the polynomial growth of $Du(B_R)$. In the bounded case, compactness of integer translates and the strong comparison principle give periodicity; the cell-problem uniqueness then finishes the proof.

The rest of the paper is organized as follows. In Section 2, we prove Theorem 1.1. In Section 3, we study the cell problem. Section 4 is devoted to periodic correctors and the proof of Theorem 1.3.

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2. THE CONSTANT-MEDIUM EQUATION

We prove Theorem 1.1 for the constant-medium equation

$$(2.1) \quad \det D^2u = \exp\{-a \cdot Du + b \cdot x - c_0\}, \quad D^2u > 0, \quad x \in \mathbb{R}^n.$$

Throughout this section, strict convexity means $D^2u > 0$.

If $a = b = 0$, then (2.1) reduces to the constant-density Monge–Ampère equation

$$\det D^2u = e^{-c_0} \quad \text{in } \mathbb{R}^n.$$

The Jörgens–Calabi–Pogorelov theorem therefore implies that every entire smooth strictly convex solution is a quadratic polynomial; see [Jör54, Cal58, Pog72]. Its constant Hessian matrix necessarily satisfies $\det D^2u = e^{-c_0}$.

Before treating the negative-coupling case, we record an elementary consequence of strict convexity.

Lemma 2.1. *Let $u \in C^\infty(\mathbb{R}^n)$ be strictly convex and assume that $Du(0) = 0$. Then*

$$u(x) \rightarrow +\infty \quad \text{as } |x| \rightarrow \infty.$$

In particular, every sublevel set of u is bounded.

Proof. For $e \in \mathbb{S}^{n-1}$, set

$$g_e(t) = u(te).$$

Then

$$g_e''(t) = e^\top D^2u(te)e > 0, \quad g_e'(0) = 0.$$

Hence $g_e'(1) > 0$ for every $e \in \mathbb{S}^{n-1}$. Since

$$e \mapsto g_e'(1) = Du(e) \cdot e$$

is continuous on the compact unit sphere,

$$\sigma := \min_{|e|=1} g_e'(1) > 0.$$

For $r \geq 1$, convexity gives

$$u(re) = g_e(r) \geq g_e(1) + (r-1)g_e'(1) \geq \min_{|e|=1} u(e) + \sigma(r-1).$$

The right-hand side tends to $+\infty$ uniformly in e . □

Proposition 2.2. *Assume that $a \neq 0$ and $a \cdot b < 0$. Then (2.1) admits no entire smooth strictly convex solution.*

Proof. This is the nonexistence theorem of Wu–Xu [WX21, Theorem 1.3]. We include the argument since the same auxiliary function will be used in the borderline case.

Suppose that a solution exists. After translating the base point, subtracting a supporting affine function, and changing the constant c_0 , we may assume

$$u(0) = 3, \quad Du(0) = 0, \quad u > 0.$$

For $C > 3$, set

$$\Omega_C = \{u < C\}, \quad \eta = C - u, \quad \kappa = 3C.$$

By Lemma 2.1, the domain Ω_C is bounded.

Define

$$Q = \exp\left(-\frac{\kappa}{\eta}\right) e^{-b \cdot x} \det D^2 u = \exp\left(-\frac{\kappa}{\eta} - a \cdot Du - c_0\right) \quad \text{in } \Omega_C.$$

The function Q extends continuously by zero to $\partial\Omega_C$. It therefore attains its maximum at an interior point $x_C \in \Omega_C$.

At x_C ,

$$(2.2) \quad 0 = (\log Q)_i = -\frac{\kappa u_i}{\eta^2} - a_\ell u_{\ell i}.$$

Contracting (2.2) with a_i and with $u^{ij}u_j$, respectively, gives

$$(2.3) \quad a^\top D^2 u a = -\frac{\kappa}{\eta^2} a \cdot Du, \quad -a \cdot Du = \frac{\kappa}{\eta^2} u^{ij} u_i u_j \geq 0.$$

At the same point,

$$\begin{aligned} 0 &\geq u^{ij} (\log Q)_{ij} \\ &= a^\top D^2 u a - a \cdot b - \frac{\kappa n}{\eta^2} - \frac{2\kappa}{\eta^3} u^{ij} u_i u_j. \end{aligned}$$

Using (2.3), we obtain

$$(2.4) \quad 0 \geq \left(\frac{\kappa}{\eta^2} - \frac{2}{\eta}\right) (-a \cdot Du) - a \cdot b - \frac{\kappa n}{\eta^2}.$$

Since $0 < u < C$ and $\kappa = 3C$,

$$\frac{\kappa}{\eta^2} - \frac{2}{\eta} = \frac{\kappa - 2\eta}{\eta^2} = \frac{C + 2u}{(C - u)^2} > 0.$$

The first term in (2.4) is therefore nonnegative. Since $a \cdot b < 0$, it follows that

$$\eta(x_C)^2 \leq \frac{\kappa n}{-a \cdot b} = \frac{3Cn}{-a \cdot b}.$$

Consequently,

$$\frac{\kappa}{\eta(x_C)} \geq c_1 \sqrt{C}$$

for some $c_1 > 0$ depending only on n and $-a \cdot b$.

Set

$$X = -a \cdot Du(x_C) \geq 0.$$

Equation (2.4) also gives

$$\left(\frac{\kappa}{\eta(x_C)^2} - \frac{2}{\eta(x_C)} \right) X \leq a \cdot b + \frac{\kappa n}{\eta(x_C)^2} < \frac{\kappa n}{\eta(x_C)^2}.$$

Thus

$$X < \frac{\kappa n}{\kappa - 2\eta(x_C)} < 3n.$$

It follows that

$$\begin{aligned} \log Q(x_C) &= -\frac{\kappa}{\eta(x_C)} - a \cdot Du(x_C) - c_0 \\ &\leq -c_1 \sqrt{C} + 3n - c_0, \end{aligned}$$

and therefore

$$Q(x_C) \rightarrow 0 \quad \text{as } C \rightarrow \infty.$$

On the other hand,

$$Q(0) = \exp \left(-\frac{3C}{C-3} - c_0 \right) \rightarrow e^{-3-c_0} > 0.$$

For sufficiently large C , this contradicts $Q(0) \leq Q(x_C)$. $\square$

Assume now that $a \cdot b = 0$. We first obtain a global oscillation estimate for $a \cdot Du$.

Lemma 2.3. *Let u solve (2.1) and suppose that $a \cdot b = 0$. Then*

$$|a \cdot Du(x) - a \cdot Du(y)| \leq 3n + 3 \quad \text{for all } x, y \in \mathbb{R}^n.$$

Proof. Fix a base point and perform an affine normalization so that

$$u(0) = 3, \quad Du(0) = 0, \quad u > 0.$$

For $C > 3$, use the same quantities

$$\Omega_C = \{u < C\}, \quad \eta = C - u, \quad \kappa = 3C,$$

and the same auxiliary function Q as in the proof of Proposition 2.2. Let x_C be its interior maximum point.

Equations (2.3) and (2.4) remain valid. Since $a \cdot b = 0$, we obtain

$$0 \geq \left(\frac{\kappa}{\eta^2} - \frac{2}{\eta} \right) (-a \cdot Du) - \frac{\kappa n}{\eta^2}$$

at x_C . Therefore

$$-a \cdot Du(x_C) \leq \frac{\kappa n}{\kappa - 2\eta(x_C)} < 3n.$$

For any fixed $x \in \mathbb{R}^n$, once $C > u(x)$, the inequality $Q(x) \leq Q(x_C)$ gives

$$-a \cdot Du(x) \leq -a \cdot Du(x_C) + \frac{\kappa}{C - u(x)} - \frac{\kappa}{\eta(x_C)}.$$

Discarding the final nonpositive term and letting $C \rightarrow \infty$, we find

$$-a \cdot Du(x) \leq 3n + 3.$$

To compare two arbitrary points, fix $x_0 \in \mathbb{R}^n$ and consider

$$v_{x_0}(y) = u(x_0 + y) - u(x_0) - Du(x_0) \cdot y + 3.$$

By convexity,

$$v_{x_0}(0) = 3, \quad Dv_{x_0}(0) = 0, \quad v_{x_0} \geq 3.$$

The function v_{x_0} satisfies an equation of the form (2.1) with the same vectors a, b , the same coupling $a \cdot b = 0$, and a possibly different constant. Applying the preceding one-sided estimate to v_{x_0} yields

$$a \cdot Du(x_0 + y) - a \cdot Du(x_0) \geq -(3n + 3).$$

Interchanging the two points gives the reverse inequality. $\square$

We use this oscillation bound in the following proposition.

Proposition 2.4. *Let $n \geq 2$, $a \neq 0$, and $a \cdot b = 0$. Suppose that $u \in C^4(\mathbb{R}^n)$ is strictly convex, solves (2.1), and satisfies*

$$\text{osc}_{\mathbb{R}^n}(a \cdot Du) < \infty.$$

Then no such solution exists.

Proof. Suppose, to the contrary, that such a solution exists. Fix a point $x_0 \in \mathbb{R}^n$ and define

$$\tilde{u}(x) = u(x + x_0) - u(x_0) - Du(x_0) \cdot x.$$

Then

$$\tilde{u}(0) = 0, \quad D\tilde{u}(0) = 0, \quad \tilde{u} \geq 0$$

by convexity. The function $\tilde{u}$ also satisfies an equation of the same form as (2.1), with the same vectors a and b , and with c_0 replaced by

$$\tilde{c}_0 = c_0 + a \cdot Du(x_0) - b \cdot x_0.$$

The condition $a \cdot b = 0$ is unchanged, and

$$a \cdot D\tilde{u}(x) = a \cdot Du(x + x_0) - a \cdot Du(x_0),$$

so the oscillation assumption is unchanged as well. Renaming $\tilde{u}$ as u , we may assume

$$u(0) = 0, \quad Du(0) = 0, \quad u \geq 0.$$

By Lemma 2.1,

$$u(x) \rightarrow +\infty \quad \text{as } |x| \rightarrow \infty.$$

Set

$$w = a \cdot Du.$$

Since $w(0) = 0$, the oscillation assumption implies

$$|w| \leq W \quad \text{in } \mathbb{R}^n$$

for some finite constant W .

Write

$$g_{ij} = u_{ij}, \quad (g^{ij}) = (g_{ij})^{-1},$$

and define the drifted linearized operator

$$\mathcal{L}f = g^{ij}f_{ij} + a_k f_k.$$

We use the summation convention and set

$$P = a_r a_s u_{rs} = a^\top D^2 u a > 0, \quad T_{ij} = a_k u_{ijk}.$$

Here and below,

$$|T|_g^2 = g^{ip} g^{jq} T_{ij} T_{pq}, \quad |\nabla P|_g^2 = g^{ij} P_i P_j.$$

Since

$$w_i = a_k u_{ki} = g_{ik} a_k,$$

we have

$$P = g^{ij} w_i w_j = |\nabla w|_g^2, \quad w_{ij} = a_k u_{kij} = T_{ij}.$$

Taking the logarithm of (2.1) gives

$$\log \det(g_{ij}) = -a \cdot Du + b \cdot x - c_0.$$

Differentiating in the x_k -direction yields

$$g^{ij} u_{ijk} = b_k - a_\ell u_{\ell k}.$$

Contracting with a_k and using $a \cdot b = 0$, we obtain

$$(2.5) \quad g^{ij} T_{ij} = -P.$$

Since

$$a \cdot Dw = P,$$

identity (2.5) also gives

$$\mathcal{L}w = g^{ij} w_{ij} + a \cdot Dw = -P + P = 0.$$

Differentiating the logarithmic equation once more gives

$$g^{ij} u_{ijrs} - g^{ip} g^{jq} u_{ijr} u_{pqs} = -a_k u_{krs}.$$

Multiplying by $a_r a_s$, we find

$$g^{ij} P_{ij} - g^{ip} g^{jq} T_{ij} T_{pq} = -a_k P_k.$$

Hence the first-order term in $\mathcal{L}$ cancels the right-hand side, and

$$(2.6) \quad \mathcal{L}P = g^{ip} g^{jq} T_{ij} T_{pq} = |T|_g^2.$$

Differentiating the definition of P also gives

$$(2.7) \quad P_i = a_r a_s u_{rsi} = T_{ij} a_j.$$

At a fixed point, choose a g -orthonormal basis whose first vector is parallel to a . In this basis,

$$g_{ij} = \delta_{ij}, \quad a = \sqrt{P} e_1.$$

This choice is pointwise; no derivatives of the frame are taken. Equations (2.6), (2.5), and (2.7) become

$$\mathcal{L}P = |T|^2, \quad \text{tr } T = -P,$$

and

$$(2.8) \quad |\nabla P|_g^2 = P \sum_i T_{1i}^2, \quad a \cdot DP = PT_{11}.$$

Choose

$$\theta = \frac{1}{4(n-1)}.$$

In particular,

$$0 < \theta < 1.$$

The chain rule, (2.6), and (2.8) give

$$\begin{aligned} \mathcal{L}(-P^{-\theta}) &= \theta P^{-\theta-1} \mathcal{L}P - \theta(\theta+1) P^{-\theta-2} |\nabla P|_g^2 \\ &= \theta P^{-\theta-1} \left(|T|^2 - (\theta+1) \sum_i T_{1i}^2 \right). \end{aligned}$$

Write

$$T = \begin{pmatrix} \tau & v^\top \\ v & B \end{pmatrix}, \quad \tau = T_{11},$$

where

$$v = (T_{12}, \dots, T_{1n})$$

and

$$B = (T_{\alpha\beta})_{2 \leq \alpha, \beta \leq n}.$$

Then

$$|T|^2 = \tau^2 + 2|v|^2 + |B|^2$$

and

$$\sum_i T_{1i}^2 = \tau^2 + |v|^2.$$

Consequently,

$$\begin{aligned} |T|^2 - (\theta+1) \sum_i T_{1i}^2 &= -\theta\tau^2 + (1-\theta)|v|^2 + |B|^2 \\ &\geq -\theta\tau^2 + |B|^2. \end{aligned}$$

By (2.5),

$$\operatorname{tr} B = -P - \tau.$$

Since B is an $(n-1) \times (n-1)$ symmetric matrix, the Cauchy–Schwarz inequality gives

$$|B|^2 \geq \frac{(\operatorname{tr} B)^2}{n-1} = \frac{(P+\tau)^2}{n-1}.$$

Thus

$$(2.9) \quad \mathcal{L}(-P^{-\theta}) \geq \theta P^{-\theta-1} \left(\frac{(P+\tau)^2}{n-1} - \theta\tau^2 \right).$$

For $\varepsilon > 0$, define

$$F_\varepsilon = -P^{-\theta} - \varepsilon u.$$

Since $P > 0$, one has $-P^{-\theta} \leq 0$. Together with $u(x) \rightarrow +\infty$, this gives

$$F_\varepsilon(x) \rightarrow -\infty \quad \text{as } |x| \rightarrow \infty.$$

Thus F_ε attains a global maximum at some point x_ε .

At x_ε ,

$$DF_\varepsilon = 0,$$

and hence

$$\theta P^{-\theta-1} DP = \varepsilon Du.$$

Choose at x_ε the adapted g -orthonormal basis used above. Contracting the preceding identity with a and using (2.8), we obtain

$$\theta P^{-\theta-1}(P\tau) = \varepsilon w.$$

Thus

$$(2.10) \quad \tau = \frac{\varepsilon}{\theta} w P^\theta, \quad |\tau| \leq \frac{\varepsilon W}{\theta} P^\theta.$$

All quantities in the rest of the argument are evaluated at x_ε .

Suppose first that

$$|\tau| \geq \frac{P}{2}.$$

Then (2.10) gives

$$\frac{P}{2} \leq \frac{\varepsilon W}{\theta} P^\theta,$$

and hence

$$(2.11) \quad P^{1-\theta} \leq \frac{2W}{\theta} \varepsilon.$$

Suppose instead that

$$|\tau| < \frac{P}{2}.$$

Then

$$P + \tau \geq \frac{P}{2}$$

and

$$\tau^2 < \frac{P^2}{4}.$$

Therefore

$$\begin{aligned} \frac{(P + \tau)^2}{n-1} - \theta \tau^2 &\geq \frac{P^2}{4(n-1)} - \frac{\theta P^2}{4} \\ &= \frac{P^2}{4} \left(\frac{1}{n-1} - \theta \right). \end{aligned}$$

Since

$$\theta = \frac{1}{4(n-1)},$$

the right-hand side equals

$$\frac{3P^2}{16(n-1)} \geq \frac{P^2}{8(n-1)}.$$

Estimate (2.9) gives

$$\mathcal{L}(-P^{-\theta}) \geq \frac{\theta}{8(n-1)} P^{1-\theta}.$$

At the maximum point,

$$DF_\varepsilon(x_\varepsilon) = 0$$

and

$$D^2F_\varepsilon(x_\varepsilon) \leq 0.$$

The first-order part of $\mathcal{L}$ therefore vanishes there, and

$$\mathcal{L}F_\varepsilon(x_\varepsilon) = g^{ij}(F_\varepsilon)_{ij}(x_\varepsilon) \leq 0.$$

At the same point,

$$\mathcal{L}u = g^{ij}u_{ij} + a \cdot Du = n + w \leq n + W.$$

Hence

$$\begin{aligned} 0 &\geq \mathcal{L}F_\varepsilon = \mathcal{L}(-P^{-\theta}) - \varepsilon \mathcal{L}u \\ &\geq \frac{\theta}{8(n-1)} P^{1-\theta} - \varepsilon(n + W). \end{aligned}$$

Hence

$$(2.12) \quad P^{1-\theta} \leq \frac{8(n-1)(n+W)}{\theta} \varepsilon.$$

Combining (2.11) and (2.12), we obtain

$$P(x_\varepsilon)^{1-\theta} \leq C\varepsilon,$$

where C is independent of ε . Since $1 - \theta > 0$, this implies

$$(2.13) \quad P(x_\varepsilon) \rightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

On the other hand, maximality of F_ε , together with $u(0) = 0$ and $u \geq 0$, gives

$$-P(x_\varepsilon)^{-\theta} \geq -P(x_\varepsilon)^{-\theta} - \varepsilon u(x_\varepsilon) = F_\varepsilon(x_\varepsilon) \geq F_\varepsilon(0) = -P(0)^{-\theta}.$$

Thus

$$P(x_\varepsilon) \geq P(0) > 0.$$

This contradicts (2.13). □

Remark 2.5. When $b = 0$, Proposition 2.4 gives another proof of the Li–Xu rigidity theorem [LX09] in the smooth strictly convex case. The quantity P used above is directly related to their affine invariant. Indeed, let $f = u^*$. Then

$$\det D^2f = e^{a \cdot y + c_0}.$$

If

$$\rho = (\det D^2f)^{-1/(n+2)}, \quad \Phi = \frac{f^{ij} \rho_i \rho_j}{\rho^2},$$

as in [LX09], then at the dual point $y = Du(x)$,

$$\Phi(y) = \frac{1}{(n+2)^2} a^\top D^2 u(x) a = \frac{P(x)}{(n+2)^2}.$$

Thus, up to a constant factor, P is the same scalar quantity in the original variables. In the null-coupling case it satisfies

$$\mathcal{L}w = 0, \quad P = |\nabla w|_{D^2 u}^2, \quad \mathcal{L}P = |T|_{D^2 u}^2, \quad \text{tr}_{D^2 u} T = -P,$$

which leads to the maximum-principle argument above.

Remark 2.6. The mixed null case $a \neq 0$, $b \neq 0$, $a \cdot b = 0$ is not reduced to the pure gradient-drift case by the natural changes of variables preserving entire convex gradient graphs. If $S \in GL(n, \mathbb{R})$ and

$$v(z) = u(Sz) + q \cdot z + d,$$

then the drift vectors become

$$a' = S^{-1}a, \quad b' = S^\top b, \quad a' \cdot b' = a \cdot b.$$

Hence an invertible change of the base variables cannot eliminate a nonzero spatial drift. The Legendre transform exchanges the two drift vectors, but it is defined on $Du(\mathbb{R}^n)$, which need not be all of $\mathbb{R}^n$. Therefore it does not in general preserve the class of entire solutions considered here.

Lemma 2.3 and Proposition 2.4 give the null-coupling obstruction.

Proposition 2.7. *Assume that*

$$a \neq 0, \quad a \cdot b = 0.$$

Then (2.1) admits no entire smooth strictly convex solution.

Proof. Assume first that $n \geq 2$. Lemma 2.3 gives

$$\text{osc}_{\mathbb{R}^n}(a \cdot Du) < \infty.$$

The conclusion therefore follows from Proposition 2.4.

If $n = 1$, then $a \neq 0$ and $ab = 0$ imply $b = 0$. The equation is

$$u'' = e^{-au' - c_0}.$$

Consequently,

$$(e^{au'})' = ae^{-c_0} \neq 0.$$

Thus $e^{au'}$ is a nonconstant affine function of x , which cannot remain positive on all of $\mathbb{R}$. This is a contradiction. $\square$

It remains to construct non-quadratic entire solutions in the two solvable regimes.

Suppose first that $a = 0$ and $b \neq 0$. After an orthogonal change of coordinates, write

$$b = \beta e_n, \quad \beta = |b| > 0.$$

Then

$$u(x', x_n) = \frac{1}{2}|x'|^2 + \frac{e^{-c_0}}{\beta^2} e^{\beta x_n}$$

is smooth and strictly convex, and

$$\det D^2 u = e^{\beta x_n - c_0} = e^{b \cdot x - c_0}.$$

It is plainly non-quadratic.

Assume now that

$$a \cdot b > 0.$$

Choose a symmetric positive definite matrix A satisfying

$$Aa = b.$$

For $n = 1$, simply take $A = b/a > 0$. For $n \geq 2$, use an orthonormal basis whose first vector is $e_1 = a/|a|$, and write

$$\frac{b}{|a|} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha = \frac{a \cdot b}{|a|^2} > 0, \quad \beta \in \mathbb{R}^{n-1}.$$

For any

$$\lambda > \frac{|\beta|^2}{\alpha},$$

the matrix

$$A = \begin{pmatrix} \alpha & \beta^\top \\ \beta & \lambda I_{n-1} \end{pmatrix}$$

is positive definite by the Schur complement criterion and satisfies $Aa = b$.

First note that the condition $Aa = b$ also gives quadratic solutions. Since $a \neq 0$, choose $\ell_0 \in \mathbb{R}^n$ so that

$$a \cdot \ell_0 = -c_0 - \log \det A.$$

Then

$$u_0(x) = \frac{1}{2} x^\top A x + \ell_0 \cdot x$$

solves (2.1). To show that the positive-coupling regime also contains non-quadratic solutions, choose $\ell \in \mathbb{R}^n$ such that

$$\gamma = a \cdot \ell \neq 0,$$

and set

$$\mu = \ell^\top A^{-1} \ell > 0, \quad K_0 = \frac{e^{-c_0}}{\det A}.$$

Fix $C_0 > 0$, and define a smooth function H on $\mathbb{R}$, up to an additive constant, by

$$e^{\gamma H'(t)} = K_0 + C_0 e^{-\gamma t/\mu}.$$

Differentiating gives

$$H''(t) = -\frac{1}{\mu} \frac{C_0 e^{-\gamma t/\mu}}{K_0 + C_0 e^{-\gamma t/\mu}},$$

and hence

$$(2.14) \quad 1 + \mu H''(t) = \frac{K_0}{K_0 + C_0 e^{-\gamma t/\mu}} = K_0 e^{-\gamma H'(t)} > 0.$$

Define

$$u(x) = \frac{1}{2} x^\top A x + H(\ell \cdot x).$$

Then

$$D^2 u = A + H''(\ell \cdot x) \ell \otimes \ell.$$

Since $A > 0$, the rank-one update criterion and (2.14) imply that $D^2 u > 0$. The matrix determinant lemma gives

$$\det D^2 u = \det A (1 + \mu H''(\ell \cdot x)).$$

Since A is symmetric and $Aa = b$,

$$\begin{aligned} -a \cdot Du + b \cdot x &= -a \cdot (Ax + H'(\ell \cdot x)\ell) + b \cdot x \\ &= -\gamma H'(\ell \cdot x). \end{aligned}$$

Using (2.14) and the definition of K_0 , we conclude that

$$\begin{aligned} \det D^2 u &= \det A K_0 e^{-\gamma H'(\ell \cdot x)} \\ &= \exp\{-a \cdot Du + b \cdot x - c_0\}. \end{aligned}$$

Because $C_0 > 0$, the function H is not quadratic, and neither is u .

Remark 2.8. Notice that the positive-coupling construction cannot remain uniformly convex as $a \cdot b \downarrow 0$ with $a \neq 0$. Indeed, every positive definite matrix A satisfying $Aa = b$ obeys

$$a^\top Aa = a \cdot b, \quad \lambda_{\min}(A) \leq \frac{a \cdot b}{|a|^2}.$$

Thus the quadratic background degenerates as the null threshold is approached.

Proof of Theorem 1.1. If $a = b = 0$, the classical Jörgens–Calabi–Pogorelov theorem gives part (i).

Suppose that $a \neq 0$ and $a \cdot b \leq 0$. If $a \cdot b < 0$, Proposition 2.2 gives nonexistence. If $a \cdot b = 0$, Proposition 2.7 gives nonexistence. This proves part (ii).

The two explicit constructions above prove part (iii) and complete the proof. $\square$

3. THE PERIODIC CELL PROBLEM

Throughout this section, $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ is normalized to have volume one. Let

$$A = A^\top > 0, \quad a \in \mathbb{R}^n, \quad V \in C^\infty(\mathbb{T}^n).$$

We prove Theorem 1.2, namely the existence and uniqueness of a constant $c_A \in \mathbb{R}$ and a zero-average function $\psi_A \in C^\infty(\mathbb{T}^n)$ satisfying

$$(3.1) \quad \det(A + D^2 \psi_A) = \exp\{-a \cdot D\psi_A + V - c_A\}, \quad A + D^2 \psi_A > 0 \quad \text{on } \mathbb{T}^n.$$

We write $c_A = c(A; a, V)$ when the dependence on the other data is relevant.

Consider the continuity path

$$(3.2) \quad \log \det(A + D^2\psi) + t a \cdot D\psi - tV + c = 0, \quad t \in [0, 1].$$

3.1. Comparison and uniqueness. We use the following comparison principle.

Lemma 3.1 (Strong comparison principle). *Let $\Omega \subset \mathbb{R}^n$ be connected, let $N = N^\top$ be a smooth matrix field on Ω , and let $u_1, u_2 \in C^2(\Omega)$ satisfy*

$$N + D^2u_i > 0, \quad i = 1, 2.$$

Assume that

$$\log \det(N + D^2u_1) + a \cdot Du_1 = \log \det(N + D^2u_2) + a \cdot Du_2 \quad \text{in } \Omega.$$

If

$$u_1 \leq u_2 \quad \text{in } \Omega$$

and equality holds at an interior point, then

$$u_1 \equiv u_2 \quad \text{in } \Omega.$$

Proof. Set

$$w = u_1 - u_2.$$

Subtracting the two equations and integrating the logarithmic determinant along the segment $u_2 + sw$, $s \in [0, 1]$, gives

$$B^{ij}w_{ij} + a \cdot Dw = 0,$$

where

$$B^{ij}(x) = \int_0^1 (N + D^2u_2 + sD^2w)_{ij}^{-1} ds.$$

The positive definite cone is convex, so every matrix inside the integral is positive definite. Thus the resulting linear operator is uniformly elliptic on compact subsets of Ω . Since $w \leq 0$ and w attains an interior maximum, the strong maximum principle gives $w \equiv 0$. $\square$

Lemma 3.2. *Suppose that (ψ_1, c_1) and (ψ_2, c_2) are two smooth solutions of (3.1). Then $c_1 = c_2$, and $\psi_1 - \psi_2$ is constant. In particular, if both functions have zero average, then $\psi_1 = \psi_2$.*

Proof. Let

$$w = \psi_1 - \psi_2.$$

At a maximum point x_+ of w ,

$$Dw(x_+) = 0, \quad D^2w(x_+) \leq 0.$$

Hence

$$A + D^2\psi_1(x_+) \leq A + D^2\psi_2(x_+).$$

The determinant is monotone on the positive definite cone, and $D\psi_1(x_+) = D\psi_2(x_+)$. Evaluating the equations at x_+ therefore gives $c_1 \geq c_2$. Applying the same argument at a minimum point of w gives $c_1 \leq c_2$. Hence $c_1 = c_2$.

Let

$$m = \max_{\mathbb{T}^n}(\psi_1 - \psi_2).$$

The equation is invariant under adding constants to ψ , so $\psi_2 + m$ satisfies the same equation as ψ_2 , and

$$\psi_1 \leq \psi_2 + m$$

and equality holds at an interior point of the periodic lift to $\mathbb{R}^n$. Lemma 3.1 gives

$$\psi_1 \equiv \psi_2 + m.$$

If both functions have zero average, then $m = 0$. □

We record some elementary properties of the normalizing constant.

Lemma 3.3. *The following statements hold.*

(i) *Suppose that (ψ, c) solves*

$$\det(A + D^2\psi) = e^{-ta \cdot D\psi + tV - c}, \quad A + D^2\psi > 0,$$

for some $t \in [0, 1]$. Then

$$(3.3) \quad t \min_{\mathbb{T}^n} V - \log \det A \leq c \leq t \max_{\mathbb{T}^n} V - \log \det A.$$

(ii) *For fixed A and a , suppose that (ψ_i, c_i) solves the cell problem with potential V_i , $i = 1, 2$. Then*

$$(3.4) \quad \min_{\mathbb{T}^n}(V_1 - V_2) \leq c_1 - c_2 \leq \max_{\mathbb{T}^n}(V_1 - V_2).$$

In particular,

$$(3.5) \quad |c_1 - c_2| \leq \|V_1 - V_2\|_{L^\infty(\mathbb{T}^n)}.$$

(iii) *Fix a and V . Suppose that $A_1, A_2 > 0$ are symmetric and*

$$A_1 \leq A_2$$

in the sense of quadratic forms. If c_{A_i} denotes the corresponding normalizing constant, then

$$c_{A_1} \geq c_{A_2}.$$

If $A_2 - A_1 > 0$, then the inequality is strict.

Proof. Let x_+ be a maximum point of ψ . Then

$$D\psi(x_+) = 0, \quad D^2\psi(x_+) \leq 0.$$

Thus

$$A + D^2\psi(x_+) \leq A,$$

and hence

$$e^{tV(x_+) - c} = \det(A + D^2\psi(x_+)) \leq \det A.$$

Therefore

$$c \geq tV(x_+) - \log \det A \geq t \min_{\mathbb{T}^n} V - \log \det A.$$

The upper bound follows similarly by evaluating at a minimum point of ψ , where $D^2\psi \geq 0$. This proves (3.3).

For the second statement, let x_+ be a maximum point of $\psi_1 - \psi_2$. At x_+ ,

$$D\psi_1 = D\psi_2, \quad D^2\psi_1 \leq D^2\psi_2.$$

It follows that

$$V_1(x_+) - c_1 \leq V_2(x_+) - c_2,$$

and hence

$$c_1 - c_2 \geq V_1(x_+) - V_2(x_+) \geq \min_{\mathbb{T}^n} (V_1 - V_2).$$

Applying the same argument at a minimum point of $\psi_1 - \psi_2$ gives the opposite bound in (3.4). Estimate (3.5) follows immediately.

Finally, let ψ_i be the solution corresponding to A_i , and let x_+ be a maximum point of $\psi_1 - \psi_2$. Then

$$D^2\psi_1(x_+) \leq D^2\psi_2(x_+),$$

so

$$A_1 + D^2\psi_1(x_+) \leq A_2 + D^2\psi_2(x_+).$$

Comparing the two equations at x_+ gives $c_{A_1} \geq c_{A_2}$. If $A_2 - A_1 > 0$, the matrix inequality at x_+ is strict, and the strict monotonicity of the determinant on the positive definite cone gives $c_{A_1} > c_{A_2}$. $\square$

For fixed A and a , we write $c_A[V]$ for the normalizing constant associated with the potential V . Lemma 3.3 also gives

$$V_1 \leq V_2 \implies c_A[V_1] \leq c_A[V_2],$$

and

$$c_A[V + \lambda] = c_A[V] + \lambda \quad \text{for every constant } \lambda \in \mathbb{R}.$$

3.2. A priori estimates. For $t \in [0, 1]$, suppose that (ψ, c) is a smooth solution of (3.2), normalized by

$$\int_{\mathbb{T}^n} \psi = 0.$$

Set

$$M = A + D^2\psi.$$

All estimates below are independent of t . The first one follows from periodicity and semiconvexity.

Lemma 3.4. *There exists a constant $C = C(A, n)$ such that every solution along the continuity path satisfies*

$$(3.6) \quad \|D\psi\|_{L^\infty(\mathbb{T}^n)} \leq C$$

and

$$(3.7) \quad \|\psi\|_{L^\infty(\mathbb{T}^n)} \leq C.$$

Proof. Since

$$A + D^2\psi > 0,$$

we have

$$D^2\psi \geq -A$$

as quadratic forms. In particular,

$$\psi_{ii} \geq -A_{ii}.$$

Fix all variables except x_i , and let

$$f(s) = \psi(x_1, \dots, x_{i-1}, s, x_{i+1}, \dots, x_n).$$

Then f is 1-periodic and satisfies

$$f'' \geq -A_{ii}.$$

We claim that

$$\|f'\|_{L^\infty([0,1])} \leq A_{ii}.$$

Indeed, for $0 \leq s < t \leq 1$,

$$f'(t) - f'(s) \geq -A_{ii}(t - s).$$

Applying the same inequality to the periodically extended function on the interval $[t, s + 1]$ gives

$$f'(s) - f'(t) \geq -A_{ii}(s + 1 - t).$$

Hence

$$\text{osc}_{[0,1]} f' \leq A_{ii}.$$

Since

$$\int_0^1 f'(s) \, ds = 0,$$

the asserted bound follows.

Applying this argument in each coordinate direction gives the explicit bound

$$\|D\psi\|_{L^\infty(\mathbb{T}^n)} \leq \left(\sum_{i=1}^n A_{ii}^2 \right)^{1/2} \leq \sqrt{n} \lambda_{\max}(A),$$

and hence (3.6). Since ψ has zero average, its minimum is nonpositive and its maximum is nonnegative. The gradient bound therefore implies

$$\text{osc}_{\mathbb{T}^n} \psi \leq \text{diam}(\mathbb{T}^n) \|D\psi\|_{L^\infty(\mathbb{T}^n)},$$

which gives (3.7). □

We also use the following determinant identity.

Lemma 3.5. *For every smooth periodic ψ ,*

$$\int_{\mathbb{T}^n} \det(A + D^2\psi) \, dx = \det A.$$

Proof. For $s \in [0, 1]$, define

$$I(s) = \int_{\mathbb{T}^n} \det(A + sD^2\psi) \, dx.$$

Since

$$A + sD^2\psi = D^2 \left(\frac{1}{2} x^\top A x + s\psi \right),$$

its cofactor matrix is divergence-free:

$$\partial_i \operatorname{cof}(A + sD^2\psi)^{ij} = 0.$$

Therefore

$$\begin{aligned} I'(s) &= \int_{\mathbb{T}^n} \operatorname{cof}(A + sD^2\psi)^{ij} \psi_{ij} \, dx \\ &= \int_{\mathbb{T}^n} \partial_i (\operatorname{cof}(A + sD^2\psi)^{ij} \psi_j) \, dx \\ &= 0. \end{aligned}$$

Thus $I(1) = I(0) = \det A$. □

Combining (3.2) with Lemma 3.5, we obtain the exact identity

$$c = \log \left(\frac{\int_{\mathbb{T}^n} e^{-ta \cdot D\psi + tV} \, dx}{\det A} \right).$$

For the *a priori* estimates we use instead the pointwise bounds (3.3). Together with Lemma 3.4, they give

$$|c| \leq C(A, V).$$

The equation

$$\det M = e^{-ta \cdot D\psi + tV - c}$$

then yields

$$(3.8) \quad 0 < \lambda \leq \det M \leq \Lambda < \infty,$$

where λ, Λ depend only on A, a, V , and n .

Proposition 3.6 (Uniform second-derivative estimate). *There exists a constant $C = C(A, a, n, \|V\|_{C^2(\mathbb{T}^n)})$ such that every solution of (3.2) satisfies*

$$(3.9) \quad C^{-1}I \leq A + D^2\psi \leq CI \quad \text{on } \mathbb{T}^n.$$

Proof. Define the full linearized operator

$$\mathcal{L}_t = M^{ij} \partial_{ij} + t a \cdot D.$$

Fix a constant $B > 0$, to be chosen below, and consider

$$G(x, \xi) = \log M_{\xi\xi}(x) - B\psi(x), \quad (x, \xi) \in \mathbb{T}^n \times \mathbb{S}^{n-1}.$$

Let (x_0, ξ_0) be a maximum point of G . After a constant orthogonal change of coordinates, we may assume that at x_0 ,

$$\xi_0 = e_1$$

and

$$M(x_0) = \text{diag}(M_1, \dots, M_n), \quad M_1 = M_{11} = \lambda_{\max}(M(x_0)).$$

All computations below are performed at x_0 .

Differentiating (3.2) twice in the x_1 -direction gives

$$M^{ij} M_{ij,11} - M^{ip} M^{jq} M_{ij,1} M_{pq,1} + t a_k M_{11,k} = t V_{11}.$$

Since

$$M_{ij,11} = M_{11,ij},$$

we obtain

$$\begin{aligned} M^{ij}(\log M_{11})_{ij} &= \frac{M^{ip} M^{jq} M_{ij,1} M_{pq,1}}{M_{11}} \\ &\quad - \frac{M^{ij} M_{11,i} M_{11,j}}{M_{11}^2} - \frac{t a_k M_{11,k}}{M_{11}} + \frac{t V_{11}}{M_{11}}. \end{aligned}$$

At the diagonal point, the difference of the two third-order quadratic expressions is nonnegative. The full symmetry of the third derivatives gives

$$M_{11,i} = M_{1i,1}.$$

Hence

$$\begin{aligned} &\frac{M^{ip} M^{jq} M_{ij,1} M_{pq,1}}{M_{11}} - \frac{M^{ij} M_{11,i} M_{11,j}}{M_{11}^2} \\ &= \sum_{i>1} \frac{M_{1i,1}^2}{M_1^2 M_i} + \sum_{i,j>1} \frac{M_{ij,1}^2}{M_i M_j M_1} \geq 0. \end{aligned}$$

Therefore

$$M^{ij}(\log M_{11})_{ij} \geq -\frac{t a_k M_{11,k}}{M_{11}} + \frac{t V_{11}}{M_{11}}.$$

The drift part of $\mathcal{L}_t$ cancels the third-order drift term:

$$t a \cdot D \log M_{11} = \frac{t a_k M_{11,k}}{M_{11}}.$$

Thus

$$(3.10) \quad \mathcal{L}_t \log M_{11} \geq \frac{t V_{11}}{M_{11}}.$$

On the other hand,

$$\begin{aligned} \mathcal{L}_t(-B\psi) &= -B M^{ij} \psi_{ij} - B t a \cdot D\psi \\ &= -B n + B \text{tr}(M^{-1} A) - B t a \cdot D\psi, \end{aligned}$$

because

$$M^{ij} \psi_{ij} = M^{ij} (M_{ij} - A_{ij}) = n - \text{tr}(M^{-1} A).$$

Since $G(\cdot, e_1)$ has a maximum at x_0 ,

$$\mathcal{L}_t G(x_0, e_1) \leq 0.$$

Combining this with (3.10), we obtain

$$(3.11) \quad 0 \geq B \operatorname{tr}(M^{-1}A) - Bn - Bt a \cdot D\psi + \frac{tV_{11}}{M_{11}}.$$

Let

$$\alpha = \lambda_{\min}(A) > 0, \quad K_V = \sup_{x \in \mathbb{T}^n} \|D^2V(x)\|_{\text{op}}.$$

Then

$$\operatorname{tr}(M^{-1}A) \geq \alpha \operatorname{tr}(M^{-1}) \geq \frac{\alpha}{M_{11}}.$$

Hence

$$\frac{tV_{11}}{M_{11}} \geq -\frac{K_V}{\alpha} \operatorname{tr}(M^{-1}A).$$

Using the C^1 bound from Lemma 3.4, inequality (3.11) gives

$$0 \geq \left(B - \frac{K_V}{\alpha}\right) \operatorname{tr}(M^{-1}A) - B(n + |a| \|D\psi\|_{L^\infty}).$$

Choose

$$B = 1 + \frac{K_V}{\alpha}.$$

Then

$$\operatorname{tr}(M(x_0)^{-1}A) \leq C.$$

Since $A^{1/2}M(x_0)^{-1}A^{1/2}$ is positive definite, its operator norm is bounded by its trace.

Hence

$$A^{1/2}M(x_0)^{-1}A^{1/2} \leq CI,$$

and therefore

$$M(x_0) \geq C^{-1}A \geq C^{-1}I.$$

Combining this lower eigenvalue bound with the upper determinant bound in (3.8), we obtain

$$M_{11}(x_0) \leq C.$$

Since G attains its maximum at (x_0, e_1) , for every $(x, \xi) \in \mathbb{T}^n \times \mathbb{S}^{n-1}$,

$$\log M_{\xi\xi}(x) - B\psi(x) \leq \log M_{11}(x_0) - B\psi(x_0).$$

The C^0 estimate for ψ therefore gives

$$M_{\xi\xi}(x) \leq C.$$

Thus

$$M \leq CI \quad \text{on } \mathbb{T}^n.$$

Finally, the lower determinant bound in (3.8) and the upper eigenvalue bound imply

$$M \geq C^{-1}I.$$

This proves (3.9). □

Higher-order estimates follow from standard regularity theory.

Proposition 3.7. *For every integer $k \geq 0$, there exists a constant*

$$C_k = C_k(A, a, n, \|V\|_{C^{k+2}(\mathbb{T}^n)})$$

such that every zero-average solution of (3.2) satisfies

$$(3.12) \quad \|\psi\|_{C^k(\mathbb{T}^n)} + |c| \leq C_k.$$

Proof. Proposition 3.6 gives uniform ellipticity:

$$C^{-1}I \leq A + D^2\psi \leq CI.$$

Since $D^2\psi$ is uniformly bounded, $D\psi$ is uniformly Lipschitz. Hence the right-hand side of

$$\log \det(A + D^2\psi) = -ta \cdot D\psi + tV - c$$

is uniformly bounded in $C^\alpha(\mathbb{T}^n)$ for every fixed $\alpha \in (0, 1)$. The Evans–Krylov estimate for concave uniformly elliptic equations [Eva82, Kry83] therefore gives

$$\|\psi\|_{C^{2,\alpha}(\mathbb{T}^n)} \leq C.$$

Differentiating the equation produces uniformly elliptic linear equations for the derivatives of ψ . Standard Schauder estimates and induction yield (3.12). $\square$

3.3. The continuity method.

Proof of Theorem 1.2. Let $\mathcal{S} \subset [0, 1]$ be the set of parameters t for which there exists a smooth pair (ψ, c) solving (3.2), with $A + D^2\psi > 0$ and

$$\int_{\mathbb{T}^n} \psi = 0.$$

At $t = 0$, the pair

$$\psi = 0, \quad c = -\log \det A$$

solves (3.2). Hence

$$0 \in \mathcal{S}.$$

To prove openness, fix $t_0 \in \mathcal{S}$, and let (ψ, c) be the corresponding solution. Set

$$M = A + D^2\psi.$$

The linearization of the left-hand side of (3.2) in the variables (ψ, c) is

$$(3.13) \quad (\eta, \kappa) \mapsto L\eta + \kappa, \quad L\eta = M^{ij}\eta_{ij} + t_0 a \cdot D\eta.$$

We view this as a map

$$C_0^{2,\alpha}(\mathbb{T}^n) \times \mathbb{R} \longrightarrow C^\alpha(\mathbb{T}^n),$$

where

$$C_0^{2,\alpha}(\mathbb{T}^n) = \left\{ \eta \in C^{2,\alpha}(\mathbb{T}^n) : \int_{\mathbb{T}^n} \eta = 0 \right\}.$$

The strong maximum principle gives

$$\ker L = \{\text{constants}\}.$$

As an elliptic operator on the compact manifold $\mathbb{T}^n$, L is Fredholm as a map

$$C^{2,\alpha}(\mathbb{T}^n) \longrightarrow C^\alpha(\mathbb{T}^n).$$

Its index is zero. Indeed, the lower-order drift can first be removed through the elliptic homotopy

$$L_s \eta = M^{ij} \eta_{ij} + s t_0 a \cdot D\eta, \quad 0 \leq s \leq 1,$$

and the positive definite principal coefficients can then be deformed through elliptic operators to those of the flat Laplacian. The Fredholm index is constant along these homotopies, while the Laplacian on $\mathbb{T}^n$ has index zero. It follows that

$$\text{codim Ran } L = 1.$$

Also,

$$1 \notin \text{Ran } L.$$

Indeed, if $L\eta = 1$, then at a maximum point of η , $L\eta \leq 0$, which is impossible. Hence

$$C^\alpha(\mathbb{T}^n) = \text{Ran } L \oplus \text{span}\{1\}.$$

Adding a constant to η does not change $L\eta$, so restricting the domain of L to the zero-average subspace does not change its range. Hence the map in (3.13) is surjective.

It is also injective. If

$$L\eta + \kappa = 0$$

and η has zero average, then $\kappa > 0$ is impossible by evaluating at a minimum point of η , while $\kappa < 0$ is impossible by evaluating at a maximum point. Thus $\kappa = 0$, and the strong maximum principle gives that η is constant. Its zero average then implies $\eta = 0$.

Thus

$$(\eta, \kappa) \longmapsto L\eta + \kappa$$

is an isomorphism. The implicit function theorem proves that $\mathcal{S}$ is open.

The *a priori* estimates in Lemmas 3.4 and Proposition 3.7 imply that $\mathcal{S}$ is closed. Indeed, if

$$t_j \in \mathcal{S}, \quad t_j \rightarrow t_\infty,$$

then the corresponding solutions admit uniform C^k bounds for every k . A diagonal compactness argument yields a smooth limiting solution at t_∞ .

Thus $\mathcal{S} = [0, 1]$. At $t = 1$, we obtain a smooth pair (ψ_A, c_A) satisfying (3.1). Lemma 3.2 gives uniqueness of c_A , and uniqueness of the zero-average solution ψ_A . $\square$

4. PERIODIC CORRECTORS

Throughout this section, $A = A^\top > 0$ is fixed, and (ψ_A, c_A) denotes the unique zero-average solution of the cell problem

$$(4.1) \quad \det(A + D^2\psi_A) = \exp\{-a \cdot D\psi_A + V - c_A\}, \quad A + D^2\psi_A > 0 \quad \text{on } \mathbb{T}^n.$$

We extend V and ψ_A periodically to $\mathbb{R}^n$, and write

$$M_A = A + D^2\psi_A, \quad \Phi_A(x) = \frac{1}{2}x^\top A x + \psi_A(x).$$

Thus

$$D^2\Phi_A = M_A.$$

We study entire solutions of

$$(4.2) \quad \det D^2u = \exp\{-a \cdot Du + b \cdot x + V(x) - c_0\} \quad \text{in } \mathbb{R}^n.$$

4.1. Quadratic-periodic solutions.

Theorem 4.1. *Let $A = A^\top > 0$, $\ell \in \mathbb{R}^n$, and $\psi \in C^\infty(\mathbb{T}^n)$. Suppose that*

$$A + D^2\psi > 0.$$

Then

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi(x)$$

solves (4.2) if and only if

$$b = Aa, \quad c_0 + a \cdot \ell = c_A, \quad \psi = \psi_A + C$$

for some constant $C \in \mathbb{R}$.

In particular, if $b = Aa$, then

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi_A(x) + C$$

is a solution precisely when $c_0 + a \cdot \ell = c_A$, with $C \in \mathbb{R}$ arbitrary.

Proof. For

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi(x),$$

we have

$$Du = Ax + \ell + D\psi, \quad D^2u = A + D^2\psi.$$

Since A is symmetric,

$$-a \cdot Du + b \cdot x + V - c_0 = (b - Aa) \cdot x - a \cdot \ell - a \cdot D\psi + V - c_0.$$

Hence the equation becomes

$$(4.3) \quad \det(A + D^2\psi) = \exp\{(b - Aa) \cdot x - a \cdot D\psi + V - (c_0 + a \cdot \ell)\}.$$

The left-hand side of (4.3) is $\mathbb{Z}^n$ -periodic. The functions $D\psi$ and V are also $\mathbb{Z}^n$ -periodic. Set

$$m = b - Aa.$$

If $m \neq 0$, then some coordinate m_i is nonzero. Taking $k = e_i \in \mathbb{Z}^n$ and comparing (4.3) at x and $x + k$ gives

$$e^{m \cdot k} = 1.$$

This is impossible because $m \cdot k = m_i \neq 0$. Therefore

$$b = Aa.$$

With this compatibility imposed, (4.3) reduces to

$$\det(A + D^2\psi) = \exp\{-a \cdot D\psi + V - (c_0 + a \cdot \ell)\}.$$

The uniqueness of the cell problem, without imposing the additive normalization on ψ , gives

$$c_0 + a \cdot \ell = c_A$$

and

$$\psi = \psi_A + C$$

for some constant C . Conversely, under these conditions the full-space equation reduces to (4.1). The additive constant does not occur in the equation. $\square$

Remark 4.2. If $a \neq 0$ and $b = Aa$, then for every $c_0 \in \mathbb{R}$ one can choose $\ell \in \mathbb{R}^n$ so that

$$a \cdot \ell = c_A - c_0.$$

If $a = 0$, then $b = Aa$ is equivalent to $b = 0$, and the condition on the normalizing constant reduces to

$$c_0 = c_A.$$

4.2. The compatibility condition. We use the following elementary semiconvexity lemma.

Lemma 4.3. *Let $f \in C^2(\mathbb{R}^n)$ satisfy*

$$D^2 f \geq -\Lambda I \quad \text{in } \mathbb{R}^n$$

for some $\Lambda \geq 0$.

(i) *If $f \in L^\infty(\mathbb{R}^n)$, then*

$$(4.4) \quad \|Df\|_{L^\infty(\mathbb{R}^n)} \leq \sqrt{2\Lambda \operatorname{osc}_{\mathbb{R}^n} f}.$$

When $\Lambda = 0$, the conclusion is understood as $Df \equiv 0$.

(ii) *If*

$$(4.5) \quad \sup_{B_R} |f| = o(R^2) \quad \text{as } R \rightarrow \infty,$$

then

$$(4.6) \quad \sup_{B_R} |Df| = o(R) \quad \text{as } R \rightarrow \infty.$$

Proof. Fix $x \in \mathbb{R}^n$ and $e \in \mathbb{S}^{n-1}$, and define

$$g(t) = f(x + te).$$

Then

$$g''(t) \geq -\Lambda.$$

Suppose first that f is bounded and

$$P = g'(0) > 0.$$

For $0 \leq t \leq P/\Lambda$, when $\Lambda > 0$,

$$g'(t) \geq P - \Lambda t.$$

Therefore

$$\begin{aligned} \operatorname{osc}_{\mathbb{R}^n} f &\geq g(P/\Lambda) - g(0) \\ &\geq \int_0^{P/\Lambda} (P - \Lambda t) dt = \frac{P^2}{2\Lambda}. \end{aligned}$$

Thus

$$P \leq \sqrt{2\Lambda \operatorname{osc}_{\mathbb{R}^n} f}.$$

Applying the same argument in the direction $-e$ gives the corresponding lower bound for $g'(0)$. This proves (4.4). If $\Lambda = 0$, then f is bounded and convex, hence constant.

Assume next that (4.5) holds. Fix $\delta \in (0, 1)$. If $x \in B_R$, $e \in \mathbb{S}^{n-1}$, and $t = \delta R$, semiconvexity gives

$$f(x + te) \geq f(x) + tDf(x) \cdot e - \frac{\Lambda t^2}{2}$$

and

$$f(x - te) \geq f(x) - tDf(x) \cdot e - \frac{\Lambda t^2}{2}.$$

Since

$$x \pm te \in B_{(1+\delta)R},$$

we obtain

$$|Df(x) \cdot e| \leq \frac{2 \sup_{B_{(1+\delta)R}} |f|}{\delta R} + \frac{\Lambda \delta R}{2}.$$

Taking the supremum over $x \in B_R$ and $e \in \mathbb{S}^{n-1}$, dividing by R , and letting $R \rightarrow \infty$, we find

$$\limsup_{R \rightarrow \infty} \frac{\sup_{B_R} |Df|}{R} \leq \frac{\Lambda \delta}{2}.$$

Since $\delta > 0$ is arbitrary, this proves (4.6). $\square$

Proposition 4.4. *Let*

$$u(x) = \frac{1}{2} x^\top A x + \ell \cdot x + f(x), \quad A = A^\top > 0,$$

be a smooth strictly convex solution of (4.2). Assume that

$$\omega(R) := \sup_{B_R} |Df| = o(R) \quad \text{as } R \rightarrow \infty.$$

Then

$$b = Aa.$$

Proof. Set

$$m = b - Aa.$$

Since

$$Du = Ax + \ell + Df,$$

the equation becomes

$$\det D^2 u = \exp\{m \cdot x - a \cdot \ell - a \cdot Df + V(x) - c_0\}.$$

Suppose, for contradiction, that $m \neq 0$, and set

$$e = \frac{m}{|m|}.$$

For $R > 1$, define the spherical cap

$$E_R = \left\{ x \in B_R : e \cdot x \geq \frac{R}{2} \right\}.$$

There exists $c_n > 0$ such that

$$|E_R| \geq c_n R^n.$$

On E_R ,

$$m \cdot x \geq \frac{|m|R}{2}.$$

Since V is bounded and

$$|a \cdot Df| \leq |a|\omega(R) = o(R),$$

for all sufficiently large R we have, on E_R ,

$$\log \det D^2 u \geq \frac{|m|R}{4} - C.$$

Hence

$$(4.7) \quad \int_{B_R} \det D^2 u \, dx \geq c R^n e^{|m|R/4}.$$

Because $D^2 u > 0$, the gradient map Du is injective. Indeed,

$$(Du(x) - Du(y)) \cdot (x - y) > 0 \quad \text{for } x \neq y.$$

Since Du is injective, the area formula gives

$$(4.8) \quad \int_{B_R} \det D^2 u \, dx = |Du(B_R)|.$$

On the other hand,

$$Du(B_R) \subset AB_R + \ell + B_{\omega(R)}.$$

Since $\omega(R) = o(R)$, the set on the right is contained in a Euclidean ball of radius CR for all sufficiently large R . Hence

$$(4.9) \quad |Du(B_R)| \leq CR^n.$$

Together with (4.8), the bounds (4.7) and (4.9) give a contradiction. Hence $m = 0$, that is, $b = Aa$. $\square$

Corollary 4.5. *Let*

$$u(x) = \frac{1}{2} x^\top A x + \ell \cdot x + f(x), \quad A = A^\top > 0,$$

be a smooth strictly convex solution of (4.2).

If

$$\sup_{B_R} |f| = o(R^2) \quad \text{as } R \rightarrow \infty,$$

then

$$b = Aa.$$

Proof. The strict convexity condition gives

$$A + D^2 f > 0,$$

and hence

$$D^2 f \geq -\lambda_{\max}(A)I.$$

Lemma 4.3 gives

$$\sup_{B_R} |Df| = o(R),$$

so Proposition 4.4 applies. $\square$

4.3. Bounded correctors. We first prove a compactness lemma for integer translates.

Lemma 4.6. *Let*

$$h \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$$

satisfy

$$M_A + D^2 h > 0$$

and

$$(4.10) \quad \log \det(M_A + D^2 h) - \log \det M_A + a \cdot Dh = \delta \quad \text{in } \mathbb{R}^n$$

for some constant $\delta \in \mathbb{R}$. Then the family

$$\{h(\cdot + m) : m \in \mathbb{Z}^n\}$$

is precompact in $C_{\text{loc}}^\infty(\mathbb{R}^n)$.

Proof. Since M_A is smooth, periodic, and positive definite, there are constants

$$0 < \lambda_A \leq \Lambda_A < \infty$$

such that

$$(4.11) \quad \lambda_A I \leq M_A \leq \Lambda_A I \quad \text{on } \mathbb{R}^n.$$

The positivity of $M_A + D^2 h$ gives

$$D^2 h \geq -\Lambda_A I.$$

Lemma 4.3 and the boundedness of h then yield

$$(4.12) \quad \|Dh\|_{L^\infty(\mathbb{R}^n)} \leq L$$

for some finite constant L . Set also

$$H = \|h\|_{L^\infty(\mathbb{R}^n)}.$$

Equation (4.10) is equivalent to

$$\det(M_A + D^2 h) = e^{\delta - a \cdot Dh} \det M_A.$$

Using (4.11) and (4.12), we obtain constants

$$0 < \lambda \leq \Lambda < \infty$$

such that

$$(4.13) \quad \lambda \leq \det(M_A + D^2h) \leq \Lambda \quad \text{in } \mathbb{R}^n.$$

For $m \in \mathbb{Z}^n$, set

$$h_m(x) = h(x + m), \quad U_m(x) = \Phi_A(x) + h_m(x).$$

The periodicity of M_A implies

$$D^2U_m = M_A + D^2h_m > 0,$$

and every h_m satisfies

$$(4.14) \quad \log \det(M_A + D^2h_m) - \log \det M_A + a \cdot Dh_m = \delta.$$

Equivalently,

$$(4.15) \quad \det D^2U_m = e^{\delta - a \cdot Dh_m} \det M_A.$$

The determinant bounds in (4.13) hold uniformly for all m .

Fix $x_0 \in \mathbb{R}^n$, and define

$$\begin{aligned} q_{m,x_0}(z) &:= U_m(x_0 + z) - U_m(x_0) - DU_m(x_0) \cdot z, \\ S_m(x_0, t) &:= \{x_0 + z : q_{m,x_0}(z) < t\}. \end{aligned}$$

By (4.11),

$$\frac{\lambda_A}{2}|z|^2 \leq \Phi_A(x_0 + z) - \Phi_A(x_0) - D\Phi_A(x_0) \cdot z \leq \frac{\Lambda_A}{2}|z|^2.$$

By the definitions of H and L ,

$$|h_m(x_0 + z) - h_m(x_0) - Dh_m(x_0) \cdot z| \leq 2H + L|z|.$$

Hence

$$(4.16) \quad \frac{\lambda_A}{2}|z|^2 - L|z| - 2H \leq q_{m,x_0}(z) \leq \frac{\Lambda_A}{2}|z|^2 + L|z| + 2H.$$

All constants here are independent of m and x_0 .

Choose

$$H_0 = 4(2\Lambda_A + 2L + 2H + 1).$$

The upper bound in (4.16) gives

$$(4.17) \quad B_2(x_0) \subset S_m \left(x_0, \frac{H_0}{4} \right).$$

Choose $R_0 > 2$ so large that

$$\frac{\lambda_A}{2}r^2 - Lr - 2H > H_0 \quad \text{for every } r \geq R_0.$$

The lower bound in (4.16) then gives

$$(4.18) \quad S_m(x_0, H_0) \subset B_{R_0}(x_0).$$

Hence the sections $S_m(x_0, H_0)$ have uniform inner and outer ball bounds. Set

$$\Omega_{m,x_0} = S_m(x_0, H_0) - x_0.$$

By John's lemma, there is an affine map

$$\mathcal{A}_{m,x_0}(z) = B_{m,x_0}z + y_{m,x_0}$$

such that

$$B_1 \subset \mathcal{A}_{m,x_0}(\Omega_{m,x_0}) \subset B_n.$$

The ball inclusions (4.17) and (4.18) imply uniform bounds for B_{m,x_0} , B_{m,x_0}^{-1} , and $|\det B_{m,x_0}|^{\pm 1}$. Define on the normalized domain

$$\tilde{\Omega}_{m,x_0} = \mathcal{A}_{m,x_0}(\Omega_{m,x_0})$$

the convex function

$$\tilde{v}_{m,x_0}(y) = \frac{1}{H_0} q_{m,x_0}(\mathcal{A}_{m,x_0}^{-1}(y)) - 1.$$

Then $\tilde{v}_{m,x_0} = 0$ on $\partial\tilde{\Omega}_{m,x_0}$. Since $q_{m,x_0} \geq 0$ and $q_{m,x_0}(0) = 0$,

$$\min_{\tilde{\Omega}_{m,x_0}} \tilde{v}_{m,x_0} = \tilde{v}_{m,x_0}(\mathcal{A}_{m,x_0}(0)) = -1.$$

The normalized domains and potentials therefore satisfy the standard normalization hypotheses uniformly in m and x_0 . Moreover,

$$\det D^2\tilde{v}_{m,x_0}(y) = H_0^{-n} |\det B_{m,x_0}|^{-2} \det D^2U_m(x_0 + \mathcal{A}_{m,x_0}^{-1}(y)),$$

so the normalized Monge–Ampère densities are bounded above and below by positive constants independent of m and x_0 .

By Caffarelli's interior $C^{1,\gamma}$ estimate on normalized sections [Caf90a, Caf91] (see also [Fig17, Theorem 4.20]), there exist $\gamma \in (0, 1)$ and a uniform constant C such that

$$[DU_m]_{C^\gamma(S_m(x_0, 3H_0/4))} \leq C.$$

All these estimates are applied on fixed fractional sublevel sets of the normalized potential, so the constants are uniform in m and x_0 . Since

$$Dh_m = DU_m - D\Phi_A$$

and Φ_A is fixed and smooth, the right-hand side

$$f_m(x) = e^{\delta - a \cdot Dh_m(x)} \det M_A(x)$$

of (4.15) is uniformly bounded in $C^\gamma(S_m(x_0, 3H_0/4))$ and is uniformly bounded away from zero. After pulling the equation back by $\mathcal{A}_{m,x_0}$, the normalized right-hand sides also have uniformly bounded C^γ norms, since the linear parts of $\mathcal{A}_{m,x_0}$ and their inverses are uniformly bounded.

Caffarelli's interior $C^{2,\alpha}$ estimate for Hölder densities [Caf90b, Caf91], in the formulation [Fig17, Theorem 4.42], applied to the normalized equation and then transferred back to the original coordinates, yields some $\alpha \in (0, \gamma]$ such that

$$\|D^2U_m\|_{C^\alpha(S_m(x_0, H_0/2))} \leq C.$$

The affine changes of variables above have uniformly bounded linear parts and inverses, so the constants remain independent of m and x_0 . Since

$$B_2(x_0) \subset S_m \left(x_0, \frac{H_0}{4} \right) \subset S_m \left(x_0, \frac{H_0}{2} \right),$$

we obtain a uniform $C^{2,\alpha}$ bound on $B_2(x_0)$. Together with the determinant lower bound in (4.13), this makes

$$G_m := D^2 U_m = M_A + D^2 h_m$$

uniformly elliptic on $B_2(x_0)$.

Differentiating (4.14) in the x_k -direction gives

$$(4.19) \quad G_m^{ij} \partial_{ij} (h_m)_k + a \cdot D(h_m)_k = ((M_A^{-1})^{ij} - G_m^{ij}) \partial_k (M_A)_{ij}.$$

The coefficients G_m^{ij} are uniformly elliptic and uniformly C^α , and the right-hand side is uniformly C^α , because M_A is fixed and smooth. Interior Schauder estimates applied to (4.19) give a uniform $C^{3,\alpha}$ bound for h_m on $B_{3/2}(x_0)$. Repeated differentiation and induction then yield, for every integer $r \geq 0$,

$$\|h_m\|_{C^r(B_1(x_0))} \leq C_r,$$

where C_r depends only on r , the fixed data, and finitely many norms of the fixed smooth matrix field M_A , but not on m or x_0 . Hence the family $\{h_m\}$ is uniformly bounded in every C^r norm on compact subsets of $\mathbb{R}^n$. The Arzelà–Ascoli theorem and a diagonal argument give precompactness in $C_{\text{loc}}^\infty(\mathbb{R}^n)$. $\square$

Theorem 4.7. *Let $h \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ satisfy*

$$M_A + D^2 h > 0$$

and

$$\log \det(M_A + D^2 h) - \log \det M_A + a \cdot Dh = \delta \quad \text{in } \mathbb{R}^n$$

for some constant $\delta \in \mathbb{R}$. Then

$$\delta = 0 \quad \text{and} \quad h = \text{constant}.$$

Proof. Fix $k \in \mathbb{Z}^n$, and define

$$C_k = \sup_{x \in \mathbb{R}^n} (h(x+k) - h(x)).$$

Since h is bounded, C_k is finite. Choose a sequence $x_j \in \mathbb{R}^n$ such that

$$h(x_j + k) - h(x_j) \rightarrow C_k.$$

Write

$$x_j = m_j + y_j, \quad m_j \in \mathbb{Z}^n, \quad y_j \in [0, 1]^n.$$

After passing to a subsequence,

$$y_j \rightarrow y_\infty \in [0, 1]^n.$$

By Lemma 4.6, after passing to a further subsequence,

$$h(\cdot + m_j) \rightarrow H \quad \text{in } C_{\text{loc}}^\infty(\mathbb{R}^n),$$

where H is bounded and satisfies the same equation

$$\log \det(M_A + D^2H) - \log \det M_A + a \cdot DH = \delta.$$

For every $x \in \mathbb{R}^n$,

$$h(x + m_j + k) - h(x + m_j) \leq C_k.$$

Passing to the limit gives

$$(4.20) \quad H(x + k) - H(x) \leq C_k \quad \text{for all } x \in \mathbb{R}^n.$$

The choice of the maximizing sequence gives equality at y_∞ :

$$(4.21) \quad H(y_\infty + k) - H(y_\infty) = C_k.$$

Define

$$\tilde{H}(x) = H(x + k) - C_k.$$

By (4.20),

$$\tilde{H} \leq H,$$

and by (4.21), equality holds at y_∞ . Since $k \in \mathbb{Z}^n$ and M_A is periodic, the function $H(\cdot + k)$ satisfies the same equation as H . Subtracting the constant C_k also leaves the equation unchanged. Thus both $\tilde{H}$ and H solve

$$\log \det(M_A + D^2v) + a \cdot Dv = \log \det M_A + \delta.$$

The strong comparison principle, Lemma 3.1, yields

$$H(x + k) - C_k \equiv H(x).$$

Iterating,

$$H(x + qk) - H(x) = qC_k \quad \text{for every } q \in \mathbb{N}.$$

Since H is bounded, this is possible only if $C_k = 0$. Therefore

$$h(x + k) - h(x) \leq 0 \quad \text{for all } x.$$

Applying the same argument to $-k$ and then replacing x by $x + k$ gives

$$h(x) - h(x + k) \leq 0.$$

Hence

$$h(x + k) = h(x) \quad \text{for all } x \in \mathbb{R}^n, \quad k \in \mathbb{Z}^n.$$

Thus h is $\mathbb{Z}^n$ -periodic.

At a maximum point x_+ of h ,

$$Dh(x_+) = 0, \quad D^2h(x_+) \leq 0.$$

Therefore

$$M_A(x_+) + D^2h(x_+) \leq M_A(x_+),$$

and the equation gives $\delta \leq 0$. At a minimum point x_- ,

$$Dh(x_-) = 0, \quad D^2h(x_-) \geq 0,$$

so the same equation gives $\delta \geq 0$. Hence $\delta = 0$, and $\psi_A + h$ and ψ_A solve the same periodic cell equation:

$$\log \det(A + D^2 v) + a \cdot Dv = V - c_A.$$

The uniqueness statement from Section 3 implies that their difference h is constant. $\square$

We finally prove Theorem 1.3.

Proof of Theorem 1.3. Write

$$f(x) = u(x) - \frac{1}{2}x^\top Ax - \ell \cdot x.$$

The compatibility conclusion under the subquadratic hypothesis follows from Corollary 4.5. Assume now that $f \in L^\infty(\mathbb{R}^n)$. Since a bounded function is subquadratic, we already know that $b = Aa$. Also,

$$f \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n),$$

and

$$A + D^2 f = D^2 u > 0.$$

Set

$$h = f - \psi_A.$$

Since f is bounded and ψ_A is periodic, $h \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. Also,

$$M_A + D^2 h = A + D^2 f = D^2 u > 0.$$

Using $b = Aa$, the equation for u becomes

$$\log \det(A + D^2 f) = -a \cdot Df + V - (c_0 + a \cdot \ell).$$

The cell equation gives

$$\log \det M_A = -a \cdot D\psi_A + V - c_A.$$

Subtracting these identities and using $Dh = Df - D\psi_A$, we obtain

$$\log \det(M_A + D^2 h) - \log \det M_A + a \cdot Dh = c_A - (c_0 + a \cdot \ell).$$

Theorem 4.7 therefore implies

$$c_A - (c_0 + a \cdot \ell) = 0 \quad \text{and} \quad h = \text{constant}.$$

Thus

$$c_0 + a \cdot \ell = c_A \quad \text{and} \quad f = \psi_A + \text{constant}.$$

Equivalently,

$$u(x) = \frac{1}{2}x^\top Ax + \ell \cdot x + \psi_A(x) + C$$

for some $C \in \mathbb{R}$. $\square$

DECLARATION OF COMPETING INTEREST

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author used OpenAI’s ChatGPT to assist with literature organization and with improving the language, clarity, and structure of the manuscript. After using this tool, the author reviewed and edited the content as needed, independently verified the mathematical arguments and bibliographic information, and takes full responsibility for the content of the publication.

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